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CSA0609- Design and Analysis of Algorithms

First Palindromic String in an Array and find its completity

AIM

To write a Python program to find the first palindromic string in a given array of strings and analyze its time and space complexity.

ALGORITHM

1. Start.
2. Read the number of strings n.
3. Store the n strings in an array.
4. Traverse the array from the first string.
5. For each string, check whether it is a palindrome.
6. If the string is a palindrome, display it and stop the search.
7. If no palindrome is found after checking all strings, display "No Palindrome Found".
8. Stop.

PSEUDOCODE

```
Algorithm FirstPalindrome(arr, n)
Begin
  for i ← 0 to n-1 do
    word ← arr[i]
    if word = reverse(word) then
      return word
    end if
  end for
  return "No Palindrome Found"
End
```

PYTHON CODE

main.py

Visualize

Run

```python
def is_palindrome(s):
    return s == s[::-1]


def first_palindrome(arr):
    for word in arr:
        if is_palindrome(word):
            return word

    return "No Palindrome Found"


n = int(input("Enter number of strings: "))

arr = []

for i in range(n):
    word = input("Enter string: ")
    arr.append(word)

result = first_palindrome(arr)

print("First Palindromic String:", result)
```

OUTPUT

Output

```
Enter number of strings: 5
Enter string: abc
Enter string: hello
Enter string: madam
Enter string: level
Enter string: world
First Palindromic String: madam
```

COMPLEXITY ANALYSIS

Let:

- n = number of strings
- m = maximum length of a string

TIME COMPLEXITY

Checking whether one string is a palindrome takes O(m) time.

In the worst case, all n strings may need to be checked.

Therefore:

Worst Case = O(n × m)

Best Case = O(m)
When the first string itself is a palindrome.

Average Case = O(n × m)

Space Complexity

The reverse operation s[::-1] requires additional space proportional to the string length.

Space Complexity = O(m)

RESULT

Thus, the Python program successfully finds the first palindromic string in an array, and its time complexity is O(n × m) in the worst case with O(m) auxiliary space.